

Tater Tot Brain Gym

Teacher's Guide

A homeschool retention training tool for the Tater Tot Academy

For the teacher who's reviewing, authoring, and tuning this app's content.

What this is

Tater Tot Brain Gym is a training app for the kids at home. Each kid picks their profile and does a short focused session, usually 10 to 15 minutes. The session mixes question types (multiple choice, type-the-answer, spelling bee, math drill) and uses spaced repetition to bring back items they've missed.

It is a training tool, not a game. There are light game elements (XP, streaks, badges) to keep kids engaged, but the core purpose is retention: helping them hold onto facts and skills they should already know.

What it is good for

- Drilling math facts to fluency (addition, subtraction, multiplication)
- Spelling practice with commonly-used sight words
- Vocabulary review (synonyms, antonyms, homophones)
- Geography, basic science, grammar, parts of speech
- Memory work from any curriculum that can be expressed as question and answer (CC timeline, history dates, Latin endings, science memory work)

What it is not good for

- Teaching new material. It's for review of things already introduced.
- Open-ended writing, reading comprehension, or anything that can't be graded automatically.
- Replacing real instruction. Think of this as the practice book, not the textbook.

How it works (the pedagogy)

The retention engine is based on a simplified version of the SM-2 algorithm, the same approach used by flashcard apps like Anki. Here's what happens when a kid answers a question:

If they get it right:

- The item is scheduled to appear again later, but further out.
- First time correct: 1 day out. Second time: 3 days. Then intervals grow based on how quickly and confidently they answer.
- After about 5 correct reviews in a row with good speed, the item is marked **mastered** and drops out of active rotation. You can see mastery counts in the admin progress view.

If they get it wrong (or run out of time):

- A **teach card** shows the correct answer, context, and any mnemonic.
- The item re-queues within the same session, usually 3-5 questions later.
- It also shows up in the next session, and the interval resets.
- After two lapses on an item, it gets flagged as **struggling** so you can see it in the progress view and maybe address it in actual instruction.

Why this works: spaced repetition takes advantage of the *testing effect* (retrieving information strengthens memory more than re-reading it) and the *spacing effect* (spreading review over time beats massed practice). For facts that need to be automatic (math facts, spelling, vocabulary), this is one of the most well-supported approaches in cognitive science.

Sessions and pacing

Each kid has a target session length (default 12 minutes) and a weekly goal (default 3 sessions). You can adjust both in admin mode per kid. For our kids, I'd suggest starting with every-other-day or Monday/Wednesday/Friday. Daily is better for retention but more than most kids will stick with.

Where the content comes from

This is the most important section of this guide. You should know exactly what's in the app and where it came from before you let the kids use it.

Starter packs (what ships with the app)

The six included packs cover generic grade-level content:

Pack	Source
Subtraction Facts (Gr 2-3)	Common Core 2.OA.B.2 and 3.NBT.A.2
Multiplication Facts (Gr 3-5)	Common Core 3.OA.C.7
Synonyms, Antonyms, Homophones (Gr 2-6)	Common Core L.2.5 through L.5.5
Spelling Bee: Sight Words (Gr 2-5)	Dolch sight words (public domain, 1936)
US States and Capitals (Gr 4-6)	Public record
Basic Science Facts (Gr 2-6)	NGSS K-5 standards

Important caveat. These starter packs were drafted by Claude (an AI assistant) based on its training data, which includes a broad mix of educational materials and public standards documents. The grade-level tagging for each item is Claude's best judgment based on typical expectations. It is not pulled from any single authoritative source, because no single authoritative source exists for 'what every 3rd grader should know.'

What this means for you: please review the starter packs before relying on them. The admin UI makes this easy. Click into any pack, filter by grade, and edit or delete any items that don't match your kids' level or your teaching approach. For every item you see in the app, you can:

- Change the prompt, answer, or choices
- Change the grade-level tag
- Edit the teach card (context and mnemonic shown after a wrong answer)
- Delete the item entirely

What's NOT included

Curriculum-specific content is **not** included with the app, and it should not be. Abeka's spelling lists, phonics sequence, and math workbooks are copyrighted. Classical Conversations' cycle memory work, Latin paradigms, timeline sentences, and history sentences are copyrighted. The app has not reproduced any of this material.

If you want the kids to practice this week's Abeka spelling list or the current CC Cycle memory work, you'll author those packs yourself from your teacher materials. The admin UI supports this with a 'new pack' form and per-item editing. Packs can also be imported from JSON if you get handy with the format.

Using admin mode

Admin mode is PIN-protected. The kids shouldn't be able to change their own settings, delete progress, or edit packs.

First time

Open the app, click the small 'Admin Mode' link at the bottom of the home page. You'll be asked to set an initial PIN (4 or more digits). Pick something the kids won't guess.

The Kids page

Click 'Add kid' to create a profile. Each kid needs a unique profile name (used internally) and a display name (what shows on screen, can include emoji). Set their grade level and session length. Toggle which packs are enabled for them.

After saving, you can upload an avatar photo. Kids like this. Any JPG or PNG works.

The 'Progress' button on each kid's row opens their dashboard: overall accuracy, mastery per pack, struggling items, and recent attempts. **This is the most useful view for teaching.** Struggling items tell you what to reinforce in actual instruction, not just in the app.

The Packs page

Click any pack to see all its items. You can filter by grade level, edit individual items, or delete items that don't fit. Archive a pack to remove it from kids' rotation without losing the data.

Export any pack as JSON for backup or sharing. Import a pack from JSON by dropping a file on the Packs page. This is how you'd share packs between installs or keep a version-controlled copy.

Authoring new packs

For your Abeka and CC content, you'll want to create custom packs. The cleanest workflow is:

- Prepare your word list or question list in a text file or spreadsheet
- In admin, use Import JSON to bring in a pack (format below)
- Or create items one by one in the pack editor (slower but friendlier)

Pack JSON format:

```
{
  "id": "abeka-g2-spelling-week-3",
  "title": "Abeka Grade 2, Spelling Week 3",
  "subject": "spelling",
  "questionType": "spelling-bee",
```

```
"gradeMin": 2,  
"gradeMax": 2,  
"items": [  
  {  
    "id": "abeka-g2-sp-w3-001",  
    "gradeLevel": 2,  
    "prompt": "house",  
    "answer": "house",  
    "context": "I live in a yellow house."  
  }  
]  
}
```

Question types available: *multiple-choice* (requires a 'choices' array on each item), *type-answer* (free text, compared case-insensitively), *spelling-bee* (app reads aloud via browser speech, kid types it), *math-drill* (numeric answer).

Reading the progress dashboard

Each kid has a progress page in admin mode. Here's how to read it.

Top-level stats

- **Attempts:** total questions answered (correct or incorrect). Useful for gauging how much practice they're getting.
- **Accuracy:** percentage correct across all attempts. Ranges vary by subject. For math facts drill, 85 to 95 percent is strong. For new vocabulary, 65 to 75 percent on first encounter is normal and will rise with spaced repetition.
- **Mastered:** items that have met the mastery threshold and dropped out of active rotation. Trend this up over time.
- **Struggling:** items with 2 or more lapses, or manually flagged. These are what to address in real instruction.

Per-pack breakdown

Shows mastery progress within each enabled pack. A kid with 40 out of 45 mastered in 'Subtraction Facts' is essentially fluent. One at 3 out of 50 has barely started.

Needs attention (struggling items)

The specific items the kid keeps missing. If you see a pattern (e.g. always missing multiplication facts involving 7), that's your cue to reinforce that pattern directly, not just keep drilling.

Recent attempts

A scrolling log of every question they've answered lately. Useful for spot checks (what was she actually practicing this afternoon?) and for seeing whether difficulties cluster in certain packs or times.

Practical suggestions

Start small

Enable 2 or 3 packs per kid at first. All 6 packs enabled at once can feel scattered. Start with one math pack and one language pack, run for a week, then add more.

The first session will feel long

Every item is new on day one, so there's no spaced repetition to lean on. Accuracy will look low and the teach cards will show often. This is normal. By the 3rd or 4th session, the algorithm has a better read on what each kid knows and the experience smooths out.

Watch for disengagement

If a kid is consistently timing out on questions they clearly know, or rushing through and getting easy ones wrong, they're probably checking out mentally. That's a signal to shorten the session (edit their profile, drop session minutes) or skip a day. Over-practice tanks motivation faster than under-practice tanks retention.

Use the struggling list

Every 1 to 2 weeks, check the struggling items list for each kid. These are the places the app is identifying that repeated exposure isn't fixing. Almost always this means the concept itself isn't solid, not that they need more drill. That's where real teaching earns its keep.

Don't chase 100 percent mastery

Mastery in this app means 'consistently correct over time.' Some items will never hit mastered status because the kid outgrows the pack first, or because the item is just harder than others. That's fine. What matters is the trend, not the absolute number.

Honest limitations

A few things worth knowing upfront so you're not surprised.

- **Grade tagging is approximate.** Some items in the starter packs may be tagged a grade too high or too low for our kids specifically. Please review and adjust.
- **Spelling bee uses browser speech synthesis.** Voice quality varies by browser and operating system. On modern Chrome it's good, on older browsers it can be robotic. Not a dealbreaker but worth knowing.
- **Math drill accepts only the canonical answer.** If the answer is '12' and the kid types 'twelve', it'll be marked wrong. You can edit individual items if this becomes an issue.
- **Type-answer comparison is case-insensitive and trims whitespace, but not punctuation.** 'Salt Lake City' and 'salt lake city' both match. 'Salt-Lake-City' does not.
- **No offline sync or cloud backup.** Everything lives on the machine the app is installed on. Back up the database file regularly (instructions in README).
- **The admin PIN is just a speed bump.** It's not real security. A determined kid with Google could figure out how to reset it. It's enough to stop casual wandering, not targeted evasion.

Questions, adjustments, or suggestions? Flag them as they come up. This is a tool for our family, not a product, so we're free to change whatever isn't working.